MODEL THEORY · GENERATION 5

The Generation-5 Tumor-Organoid Model

Theory, governing equations, lineage from G4, and what each added factor is meant to test
G5 腫瘤類器官模型：理論、方程式、從 G4 的承接，以及每個新增因素的因果目的

G4 established	G5 adds
A single rigid cell coupled to a boundary-anchored, crosslinked bead-spring collagen network; Gaussian material-point contact; Bell motor-clutch loading and slippage; reaction-driven motion.	Many cell disks in one organoid, cell-cell repulsion/adhesion, many simultaneous ECM contact sites, collective versus single-cell invasion, optional leader heterogeneity, nonlinear stiffening and crosslink topology plasticity.

KEY EQUATION

$$\zeta_b \, dr_i/dt = F_i^{stretch} + F_i^{bend} + F_i^{xlink} + \Sigma_a F_{ia}^{rep} + \Sigma_a F_{ia}^{cell\to ECM}$$

The collagen equation stays the common mechanical backbone. G5 changes the number of cells and the coupling terms, not the unit system.

Scope statement

This document describes the equations actually represented in the current G5 code. It is a 2-D, coarse-grained mechanism model in μm, nN and s. It is not yet a calibrated 3-D prediction of PDAC/OSCC organoid invasion. Numerical outputs are personal simulation tests until compared with experiment.

One-sentence model logic / 一句話版本
Each surface cell grips nearby collagen and reels it inward; the opposite reaction tends to move the cell outward, while cell-cell adhesion resists separation and crosslinked collagen transmits the load.

Code: generations/g5_organoid/model.py · Theory generated 2026-09-03

1 · Lineage: what is inherited and what is genuinely new

Layer	Inherited / changed	Meaning in G5
Collagen geometry	G2/G4 bead chains + outer boundary anchoring	Scaled to a 440 μm domain, 420 fibres and an organoid-sized exclusion region.
Collagen mechanics	Inherited	Axial stretch, discrete bending, freely hinged crosslinks, compression softening and overdamped bead drag.
Cell-ECM contact	G4 concept generalized	Hard contact eligibility plus Gaussian weights, now repeated over every cell and angular sector.
Motor-clutch	Inherited from G4 when enabled	Same Bell rupture, rebinding, force-velocity feedback and independent/shared-load alternatives.
Cell representation	New	One cell becomes N rigid disks packed into an organoid.
Cell-cell mechanics	New	Short-range repulsion prevents overlap; short-range adhesion supplies coarse tissue cohesion.
Migration outcome	New	Competition among cell-ECM reaction force, cell-cell adhesion and collagen resistance creates collective or single-cell escape.
Nonlinearity / plasticity	Optional G5 additions	Tensile strain-stiffening and Bell crosslink rupture/re-welding are separate ablation switches.

The main conceptual jump

• G4 asks: how does one cell load, detach from and move through one connected collagen network?

• G5 asks: how do many such cells share one matrix while also mechanically binding one another?

• Therefore G5 must introduce organoid geometry, cell-cell forces, per-cell contact bookkeeping and collective metrics. Stiffening, leaders and plasticity are not mathematically required merely because cell number increases; they are later hypotheses tested one at a time.

> **Clean baseline**
> The clean G5 baseline is: multicellular disks + cell-cell adhesion + inherited elastic collagen + inherited clutch. Strain-stiffening, leader cells and crosslink plasticity should remain explicit switches, not hidden baseline changes.

Stage map

Stage	Cell motion	New mechanism	Primary question
G5A	Fixed	N disks + adhesion	Is the organoid cohesive and force-balanced before pulling?
G5B	Fixed	Multi-cell ECM pull	Does collective contraction increase radial collagen alignment?
G5C	Fixed	Tensile strain-stiffening	Does nonlinear recruitment extend the deformation halo?
G5D	Released	Reaction-driven motion	Does adhesion select collective versus single-cell escape?
G5E	Fixed/released	Crosslink rupture + re-weld	Does topology change preserve remodeling after unloading?

2 · Shared collagen mechanics: inherited from G2/G4

Every collagen fibre is a connected sequence of beads; visible lines join consecutive beads. Crosslinks join material points on different fibres. The outer boundary is anchored so local remodeling is not confused with rigid translation of the whole network.

KEY EQUATION

$$\zeta_b \, dr_i/dt = F_i^{s} + F_i^{b} + F_i^{x} + F_i^{rep} + F_i^{act}$$

Overdamped force balance. Drag dissipates motion; elastic terms store energy. G5 contains no inertia and no SLS element.

EQUATION

$$r_i(t+\Delta t) = r_i(t) + (\Delta t/\zeta_b)\, F_i^{total}$$

Forward-Euler update for non-fixed beads; anchored beads are reset to their initial positions.

Axial stretch and compression softening

EQUATION

$$e = \ell-\ell_0,\quad F_{i\leftarrow j}^{s} = k\, e\, d/\ell,\quad k = k_t = EA/\ell_0 \text{ if } e\geq 0;\quad k = \rho k_t \text{ if } e<0$$

Tension uses the full axial stiffness. Compression uses $\rho<1$ to represent fibre microbuckling rather than an equally stiff compression spring.

Discrete bending

EQUATION

$$\Delta\kappa = (r_a-2r_b+r_c) - \kappa_0;\quad (F_a,F_b,F_c) = k_b(-\Delta\kappa,\, 2\Delta\kappa,\, -\Delta\kappa)$$

Bending penalizes changes from each fibre's initial curvature; it does not force every fibre to be straight.

Permanent elastic crosslink

EQUATION

$$p_a=(1-\alpha_a)r_i+\alpha_a r_j,\quad f_x=k_x[(p_b-p_a)-d_{x,0}]$$

The link resists separation of the two material points but is freely hinged: it carries force without prescribing an angle between fibres.

Important distinction

Overdamped drag ζb and crosslink elasticity are not collagen stress relaxation. Drag sets how fast bead positions approach force balance; a permanent elastic network still stores the deformation and tends to recover when force is removed.

G5A

Multicellular scaffold and resting balance

Replace the single cell with N mechanically interacting disks before applying organoid-scale pulling.

Organoid geometry

• Cell centres are placed on a hexagonal lattice with pitch d_0=18 μm; each disk has radius R_c=9 μm.

• Only centres satisfying $||c_a||≤R_{org}$ are kept. The current target centre radius is 63 μm, producing a coarse 2-D organoid rather than one deformable continuum.

• Collagen is excluded from the union of all cell disks plus clearance and gap, so collagen can approach perimeter cells without passing through the organoid interior.

Pairwise cell-cell force

KEY EQUATION

$$F_{a\leftarrow b}^{cc} = \{ -k_{rep}(d_0-d)e_{ab},\ d<d_0;\ +k_{adh}s_{ab}(d-d_0)e_{ab},\ d_0\le d<d_0+r_{adh};\ 0 \text{ otherwise} \}$$

Here eab points from cell a to b. Repulsion prevents overlap; adhesion attracts separated neighbours within a finite range. sab=min(sa,sb) allows a weak-adhesion leader to weaken the pair.

Why this construction?

• At $d=d_0$ the pair force is zero, so a symmetric hexagonal packing can begin near mechanical equilibrium.

• The potential is intentionally coarse-grained. k_{adh} is an effective tissue-cohesion knob, not a direct cadherin spring constant.

• The resting gate is that the organoid neither collapses nor disperses before cell-ECM traction is introduced.

Multi-cell bead exclusion

EQUATION

$$\delta_{ia}=\max(0,R_c+c-||r_i-c_a||),\quad F_i^{rep}=\Sigma_a\ k_{rep,ECM}\delta_{ia}(r_i-c_a)/||r_i-c_a||$$

G4's single-cell surface repulsion is summed over all cell disks.

G5A determines

The multi-cell geometry, non-overlap rule, finite-range cohesion and computational representation are fixed here. G5A does not yet prove collagen realignment or invasion.

G5B

Fixed organoid pulls a shared collagen network

Hold every cell centre fixed and test whether multi-cell contraction alone reorganizes collagen.

Contact eligibility + Gaussian weighting

KEY EQUATION

$$s_{af}=\|p_{af}-c_a\|-R_c,\quad 0\le s_{af}\le w_c;\quad w_{af}=\exp[-s_{af}^2/\sigma_c^2] \,/\, \Sigma_{g\in C(a)}\exp[-s_{ag}^2/\sigma_c^2]$$

Only fibres in the finite contact shell C(a) are eligible. Gaussian distance controls how an individual cell divides its prescribed total traction among those eligible fibres.

EQUATION

$$F_{af}^{cell\rightarrow ECM} = F_a w_{af}(c_a-p_{af})/\|c_a-p_{af}\|$$

The cell reels the contacted collagen material point inward. Linear shape functions distribute this point force to the two host beads: $(1-\alpha)F$ and αF.

What moves an indirect fibre?

- A directly contacted fibre receives the active term above.
- A non-contact fibre receives no second Gaussian pull. It can move only because permanent crosslinks transmit force from a directly loaded fibre, after which its own stretching and bending redistribute that force.
- Therefore a no-crosslink ablation is the causal control: indirect displacement should collapse when links are removed while geometry and direct forces are held fixed.

Radial-alignment readout

KEY EQUATION

$$S_{r,e}=2(t_e\cdot e_{r,e})^2-1,\quad S_r(q)=\mathrm{mean}_{e\ \text{in shell}\ q}S_{r,e}$$

+1 means radial/TACS-3-like, 0 isotropic, −1 tangential. Reporting shells separates near-field reorientation from whole-network averages.

G5B pass condition

- Near-field radial order increases during pull compared with the identical no-pull control.
- The grippable corona fibres are connected through crosslinks to the anchored outer network; floating contact fibres are rejected by the connectivity gate.
- Current simulations show a modest aster. Contraction mainly pulls the near-field corona inward; it does not automatically guarantee dramatic rotation of every fibre.

What the visualization means

Red/orange collagen segments have more radial orientation; blue segments are more tangential. Yellow marks are crosslinks. Beads/segments are the physical collagen representation; any separate displacement vectors are visualization overlays, not extra fibres.

G5C

Tensile strain-stiffening ablation

Ask whether a nonlinear collagen law recruits taut fibres and carries traction farther from the organoid.

KEY EQUATION

$$\varepsilon=e/\ell_0;\quad k_t(\varepsilon)=k_{t,0}\,\min[C_{max},\,\exp(\varepsilon/\varepsilon_{ref})]\text{ for }\varepsilon\geq0;\quad k(\varepsilon<0)=\rho k_{t,0}$$

Only tensile bonds stiffen. Compression stays soft. Current defaults use $\varepsilon_{ref}=0.06$ and cap $C_{max}=12$ as a numerical/model guard.

Mechanistic expectation

- As a bond enters tension, its stiffness increases, recruiting a load-bearing tract.
- A recruited tract can transmit force farther than a linear network of the same small-strain stiffness.
- The clean comparison is strain_stiffening=False versus True with the same network, pull, cell configuration and seed.

Interpretation of the current result

Negative/weak engagement is informative

In the current gentle-contraction regime, bond strains are often only about 1–5%, so $\exp(\varepsilon/0.06)$ remains modest. A small G5C effect does not falsify collagen strain-stiffening; it shows that the present loading/contact regime does not strongly enter the nonlinear branch.

What would validate this stage

Observable	Linear control	Stiffening hypothesis
Radial order vs distance	Decays relatively quickly	Alignment/recruitment persists farther
Displacement vs distance	Steeper spatial decay	Shallower decay after nonlinear recruitment
Bond strain distribution	No stiffness change	Taut-tail bonds cross ε_{ref} and carry more force
Ablation logic	Same seed and traction	Only tensile constitutive law changes

Evidence boundary

Steinwachs et al. and Mark et al. support strong nonlinear cell-matrix force transmission in collagen. They do not uniquely determine $\varepsilon_{ref}=0.06$ or $C_{max}=12$ for this synthetic 2-D network; those values remain sensitivity parameters until fitted.

6 · Cell-fibre motor-clutch in G5: inherited law, new multi-cell wiring

When clutch_dynamics=False, each gripping cell applies a prescribed total pull. When enabled, every cell is divided into angular contact sectors and each occupied sector contains an effective clutch bundle. G5 reuses G4's kinetics; its new work is repeating them over M cells and projecting the emergent traction into one shared network.

KEY EQUATION

$$v_{act}=v_0\ \max(0,1-F_{site}/F_{stall}),\quad dx/dt=\max(0,v_{act}-v_{sub}),\quad f_j=k_c x_j$$

Motor force-velocity feedback and relative loading. A yielding soft fibre (large substrate velocity vsub) reduces clutch extension growth.

EQUATION

$$k_{off}(f)=k_{off}^{\ 0}\ \exp(f/F_b),\quad P_{off}=1-\exp[-k_{off}(f)\Delta t],\quad P_{on}=1-\exp(-k_{on}\Delta t)$$

Bell slip-bond rupture and stochastic rebinding. Higher force accelerates detachment.

Two load-sharing hypotheses

Mode	Site traction	Failure interpretation
Independent	Fsite=Σj kcxj over bound clutches	One clutch can rupture while the remaining extensions are unchanged.
Shared load	fi=Fsite/i for i bound clutches	After one rupture, the same site load is divided among fewer clutches, increasing remaining-clutch hazard.

EQUATION

$$f_i=F_{site}/i,\quad r_i=i\ k_{off}^{\ 0}\ \exp[F_{site}/(iF_b)]$$

Equal load-sharing cluster: ri is the whole-cluster rupture hazard when i clutches remain.

G5-specific correctness rules

• M cells × angular sectors are flattened into one site index; the owning cell is retained for reaction-force bookkeeping.

• An empty sector cannot bind or load a clutch. The active-mask guard prevents phantom traction against nonexistent collagen.

• As a released cell moves, the gripped fibre is re-selected periodically. Current clutch state remains attached to the sector slot; this is an explicit approximation.

Do not confuse the two Bell laws

Bell rupture on a cell-fibre clutch produces adhesion slippage. Bell rupture on a fibre-fibre crosslink changes ECM topology. They use different parameters and answer different biological questions.

G5D

Release cells: collective invasion versus single-cell escape

Let every disk move under its own cell-ECM reaction force plus pairwise cell-cell mechanics.

KEY EQUATION

$$\zeta_c\, dc_a/dt = F_a^{reaction} + \Sigma_{b\neq a} F_{a\leftarrow b}^{cc}, \quad F_a^{reaction} = -\Sigma_{s\in a} F_s^{cell\to ECM}$$

A cell reels collagen inward; Newton's opposite reaction points toward the gripped ECM and tends to pull that cell outward. Cell drag converts net force to velocity.

EQUATION

$$v_a = F_a^{net}/\zeta_c, \quad \text{if } ||v_a||>v_{max}, \text{ set } v_a \leftarrow v_{max} v_a/||v_a||$$

The speed cap is a biological/numerical guard; it should be reported whenever it is active.

Mode selection is a force competition

Condition	Mechanical balance	Expected phenotype
High adhesion	Cell-cell restoring force competes strongly with outward reaction	Cohesive/collective front; smaller individual escape
Low adhesion	Cell-ECM reaction more easily exceeds cohesion	Dispersal and single-cell escape
No clutch	Reaction force collapses toward residual/contact effects	Much less active invasion
No crosslinks	ECM traction is poorly transmitted	Less organized load path and reduced invasion in the current baseline

Optional leader/EMT-like heterogeneity

EQUATION

$$s_a = s_{leader} < 1 \text{ for selected outer cells;} \quad k_{adh,ab} = k_{adh} \min(s_a, s_b); \quad F_a^{pull} = \lambda_a F^{pull}$$

Leaders may have weaker adhesion and/or stronger traction. In the current implementation the outermost fraction is selected deterministically; this is a modeling hypothesis, not emergent EMT.

What G5D does not yet include

Cells are rigid disks without deformation, division, nucleus, polarity field, proteolysis or 3-D confinement. Thus 'invasion' here means outward displacement driven by traction-reaction and adhesion balance, not full biological tissue invasion.

G5E

Crosslink topology plasticity

Allow loaded fibre-fibre welds to rupture and new current crossings to re-weld, creating mechanical memory.

KEY EQUATION

$$k_{off,x}(F_x)=k_{off,x}^{0}\exp(F_x/F_{b,x}),\quad P_{rupture}=1-\exp[-k_{off,x}(F_x)\Delta t_x]$$

Loaded crosslinks rupture faster. Current defaults are k0=2×10−4 s−1 and Fb,x=3 nN; these are model assumptions pending calibration.

EQUATION

$$\text{new crossing at time t: } d_{x,0}\leftarrow p_b(t)-p_a(t)=0 \text{ at the crossing;}\quad \text{accept with probability } p_{reform}$$

A re-weld is stress-free in the deformed configuration. That changes the link topology/rest state and can preserve remodeling after unloading.

Plasticity readouts

EQUATION

$$\kappa_{topo}=1-|L_0\cap L_T|/|L_0|$$

Fraction of original crosslinks permanently replaced. Elastic control gives 0; a rewired network gives >0. This is the current clean topology-memory metric.

EQUATION

$$\kappa_{rms}=u_{residual}/u_{peak},\quad \kappa_{order}=(S_{final}-S_0)/(S_{peak}-S_0)$$

Residual-displacement and residual-order measures. In the present network κrms is confounded by slow elastic relaxation/near-zero modes and should not be used alone to claim plasticity.

Why this is distinct from SLS

- An SLS changes stress with time while keeping the same material connections.
- G5E changes which fibres are welded together and where the new stress-free state is created.
- Therefore G5E represents topology-changing remodeling, not merely viscous relaxation.

Current finding

The mechanism is stress-selective and produces rupture/reformation events. However, matched two-hour invasion distance is close to the clutch baseline, so the present simulation does not show that plasticity is the dominant short-time invasion driver.

9 · Ablation model: how to organize the effect of every added factor

An ablation is not a new biological model by itself. It is a matched comparison in which one mechanism is removed or activated while the initial network, seed, duration and other parameters are held constant.

KEY EQUATION

$$\Delta Y_{(-m)} = Y_{\text{baseline}} - Y_{\text{mechanism m removed}}$$

Y can be invasion distance, cohesion, radial order, displacement reach, traction or topology memory. The sign and size show the model-internal causal contribution of mechanism m.

Ablation	What is isolated	Primary readout
No clutch	Active cell-fibre loading	Invasion, traction, slip events
No adhesion	Organoid cohesion	Dispersion/cohesion and single-cell escape
No crosslinks	Network force transmission	Indirect displacement, radial order, invasion
Stiffening off/on	Nonlinear tensile recruitment	Alignment/displacement reach
Plasticity off/on	Topology memory	κ_{topo} and post-unload persistence
Leaders off/on	Cell heterogeneity hypothesis	Leader displacement and follower cohesion

Current matched two-hour simulation summary

Condition	Reported displacement/invasion	Interpretation boundary
Clutch baseline	~25 μm	Reference within one 2-D seed/parameter set
No clutch	~4.7 μm	Clutch is the main active-motion driver in this baseline
No adhesion	~68 μm; worse cohesion	Adhesion restrains dispersal and maintains a collective
No crosslinks	~7.7 μm	Connected collagen is important for transmitting the active load
Stiffening	~24 μm	Little effect on this outcome/loading regime
Plasticity	~25 μm	Topology changes, but short-time invasion barely changes

Do not overclaim

These ablations provide causal attribution inside the current model. They are not experimental effect sizes and do not prove the same ranking in a real tumour until replicated over seeds and calibrated to organoid movies.

10 · Parameter map: measured scale, inherited value, or explicit assumption

Parameter	Current default	Role / status
Cell radius Rc	9 µm	Single-cell scale; consistent with ~15–20 µm diameter, not a fit to one cell line
Organoid centre radius	63 µm	2-D core geometry; displayed organoid scale is larger after adding cell radius
Cell spacing d0	18 µm	Touching hexagonal disks; defines zero pair force
cc repulsion	40 nN/µm	Effective non-overlap penalty; model parameter
cc adhesion	6 nN/µm	Effective cohesion knob; not a cadherin molecular spring
Adhesion range	6 µm	Finite neighbor interaction range; model assumption
Collagen modulus	3 MPa	Softened G3/G4 choice for visible deformation; not tissue-matched calibration
Crosslink stiffness	10 nN/µm	Softened G3/G4 effective link penalty
Fibres / domain	420 / 440 µm	Organoid-scale computational network
Crosslink retain fraction	0.85	Connectivity target in this generator; not equal to G4's probability mechanism
Pull per cell	12 nN	Constant-pull branch only; coarse tens-of-nN scale
Bead timestep	0.05 s	Inherited numerical resolution
Cell drag / speed cap	400 nN·s/µm / 0.02 µm/s	Effective mobility and guard; must be calibrated

Motor-clutch defaults inherited from G4

N/site	kc	kon	koff0	Fb	v0	Fstall/site
12	2 nN/µm	0.055 s^{-1}	0.018 s^{-1}	1.5 nN	0.025 µm/s	8 nN

> **Critical provenance warning**
> G4 crosslink_probability and G5 crosslink_fraction are not the same construction. Their raw values should not be compared as though they were one calibrated biological density.

11 · What would make G5 biologically credible?

Target	Current status	Required comparison
Resting cohesion	Mechanically stable hex packing	Compare cell-neighbour distances and cluster shape before pull
Radial alignment	Near-field aster is modest	Measure Sr versus distance and time from Kolade movies
Indirect matrix motion	Crosslink-mediated by construction	No-crosslink control + force-path/connectivity analysis
Clutch slippage	Stochastic slips occur	Compare load–detach–recoil timing with experimental videos
Collective vs single-cell	Adhesion sweep creates mode differences	Quantify strand cohesion, escaped-cell fraction and speed
Long-range deformation	Current displacement/reach too small	Compare μm displacement and radial decay over ~100–200 μm
Plastic memory	κtopo detects rewiring	Load–unload experiment or topology-sensitive proxy

Main limitations to say aloud

• 2-D network: projected intersections and paths do not reproduce 3-D pore connectivity.

• Rigid disks: no cell deformation, nucleus, cortical tension or shape-dependent contact area.

• No proliferation, death, nutrient field or growth pressure.

• No proteolysis/MMP-mediated fibre removal or pore opening.

• No explicit focal-adhesion maturation, polarity field or protrusion geometry in the organoid baseline.

• Several effective coefficients are exploratory and have not been fitted to PDAC/OSCC data.

• A single seed and a single outcome cannot establish robustness; matched ensembles and sensitivity analysis are required.

Recommended interpretation

> **What G5 can claim now**
> It is a controlled 2-D mechanism platform showing how inherited clutch traction, cell-cell cohesion and crosslinked collagen jointly produce different organoid behaviors. It is not yet a predictive digital twin of tumour invasion.

12 · Literature support and evidence boundaries

1. Lee et al. (2014), PLOS ONE 9:e111896, doi:10.1371/journal.pone.0111896 — bead-spring collagen fibre and size/modulus scales.

2. Abhilash et al. (2014), Biophysical Journal 107:1829–1840, doi:10.1016/j.bpj.2014.08.029 — contractile cells remodeling connected discrete-fibre networks.

3. Wang et al. (2014), Biophysical Journal 107:2592–2603, doi:10.1016/j.bpj.2014.09.044 — tension-driven alignment and long-range force transmission.

4. Bell (1978), Science 200:618–627, doi:10.1126/science.347575 — force-accelerated slip-bond off-rate.

5. Bangasser & Odde (2013), Cell. Mol. Bioeng. 6:449–459, doi:10.1007/s12195-013-0296-5 — stochastic motor-clutch loading framework.

6. Adebowale et al. (2021), Nature Materials 20:1290–1299, doi:10.1038/s41563-021-00981-w — motor-clutch-related effective parameter scale; not evidence for SLS collagen bonds here.

7. Erdmann & Schwarz (2004), arXiv:cond-mat/0403552 — equal-load-sharing adhesion-cluster birth/death hypothesis.

8. Ilina & Friedl (2020), Nature Cell Biology, doi:10.1038/s41556-020-0552-6 — qualitative basis for cell-cell cohesion/jamming as a collective-versus-single-cell control.

9. Steinwachs et al. (2016), Nature Methods 13:171–176, doi:10.1038/nmeth.3685 — cell forces in nonlinear biopolymer networks.

10. Mark et al. (2020), eLife 9:e51912 — organoid-scale nonlinear collagen deformation and long-range force transmission targets.

11. Nam et al. (2016), Biophysical Journal 111:2296–2308, doi:10.1016/j.bpj.2016.10.002 — viscoplastic remodeling and load–unload metrics.

12. Wisdom et al. (2018), Nature Communications, doi:10.1038/s41467-018-06641-z — matrix plasticity as a control of confined migration.

13. Kim/Ahn et al. (2019), Acta Biomaterialia 84:280, PMID 30500449 — HNSCC spheroid contraction precedes radial collagen alignment and invasion.

14. Carey et al. (2016), Integrative Biology 8:821–835, doi:10.1039/C6IB00030D — collagen alignment and cell protrusion/migration feedback.

> Evidence rule
>
> The papers support model structure, qualitative mechanism and order-of-magnitude targets. They do not convert current coarse-grained coefficients into fitted PDAC/OSCC values. Every numerical result must remain labeled as simulation output until experimentally calibrated.

Repository sources used for this theory

- generations/g5_organoid/model.py and README.md
- generations/g5_organoid/parameter_provenance.md
- docs/G5_validation_targets.md and docs/G5_research_findings.md
- CITATIONS.md and docs/g4_parameter_provenance.md